

Product Carbon Footprint Study Report

Product Name: diammonium phosphate

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: diammonium phosphate

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg diammonium phosphate

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg diammonium phosphate.

2. System Boundary

diammonium phosphate production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of diammonium phosphate produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.955 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

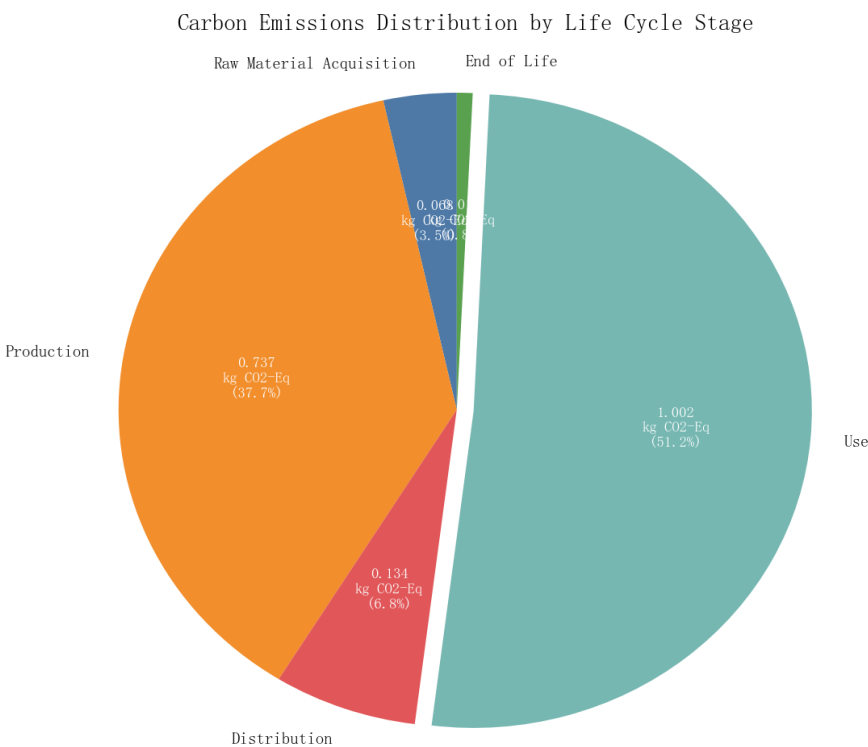


Table 1: Carbon Emissions by Life Cycle Stage

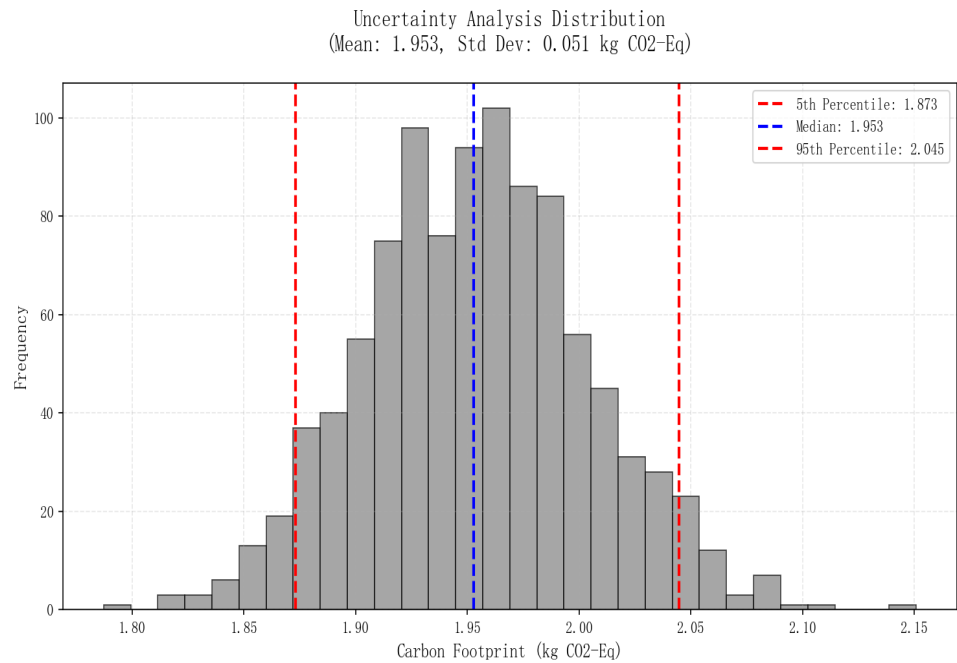
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.068	3.49
Production	0.737	37.69
Distribution	0.134	6.84

Use	1.002	51.23
End of Life	0.015	0.75
Total	1.955	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	1.953
Standard Deviation	0.051
Coefficient of Variation (CV)	2.62%
Median	1.953
5th Percentile	1.873
95th Percentile	2.045

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the diammonium phosphate product is calculated at 1.955 kg CO₂-Eq per functional unit, which reflects a moderate level of greenhouse gas emissions typical for phosphate-based fertilizer production. The uncertainty analysis supports this result with a mean value of 1.953 kg CO₂-Eq, a low coefficient of variation of 2.62%, and a narrow 90% confidence interval (1.873–2.045 kg CO₂-Eq), indicating high reliability in the assessment. This stability suggests that the LCA model captures the core emission patterns effectively.

Analysis of lifecycle stage contributions reveals that the “use” phase dominates the carbon footprint, contributing 1.002 kg CO₂-Eq (51.2%), likely due to direct emissions from soil application, such as nitrous oxide (N₂O) release during fertilizer degradation. The “production” stage is the second largest contributor at 0.737 kg CO₂-Eq (37.7%), reflecting energy-intensive processes like ammonia synthesis and phosphoric acid production. The “distribution” stage accounts for 6.8% (0.134 kg CO₂-Eq), while “raw material” extraction and “end-of-life” contribute only 3.5% and 0.8%, respectively. This distribution highlights that emission reduction efforts must prioritize use-phase management and production efficiency.

Notably, all top process contributions are listed as “Unknown,” indicating a gap in detailed process-level reporting. This suggests that the current assessment relies on aggregated or secondary data, limiting the ability to pinpoint specific emission hotspots within production or raw material supply chains. Clarifying these unknowns would enhance the actionability of the findings.

2. Statement of Assumptions and Limitations

This LCA study relies on several key assumptions that influence the results. A primary assumption is that the “use” phase emissions are modeled based on default emission factors for nitrogen fertilizers, which assume a direct release of N₂O at a fixed rate per kg of nitrogen applied. This approach may not reflect variable field conditions, such as soil type, moisture, or crop management practices, leading to potential over- or underestimation of actual on-site emissions. Additionally, the “production” stage likely assumes average energy mixes for ammonia and phosphoric acid manufacturing, without regional differentiation.

Limitations in data sources and quality are evident, as all top process contributions are listed as “Unknown.” This implies that the LCA model utilized secondary data from generic databases (e.g., Ecoinvent or EF database) rather than primary data from the specific producer. The lack of supplier-specific information for raw materials and energy usage reduces accuracy. Furthermore, the narrow confidence interval suggests low variability but does not account for systematic biases due to missing data on newer, low-carbon technologies.

The system boundary is restricted to a cradle-to-grave scope, covering raw material extraction, production, distribution, use, and end-of-life. However, it excludes capital goods, infrastructure, and biogenic carbon sequestration, which may be relevant for organic amendments. The exclusion of application-specific factors (e.g., tillage methods) further limits the model's representativeness for different agricultural contexts. No cut-off criteria or recycling credits were reported, which could affect comparative assessments.

3. Suggestions for Improvement

- ****Prioritize use-phase emission reduction****: Since the use stage contributes over 50% of the total carbon footprint, adopt nitrification inhibitors or enhanced-efficiency fertilizers to reduce N₂O emissions by up to 25–50%. Promote precision agriculture techniques (e.g., variable-rate application) to optimize nutrient use efficiency and minimize excess fertilizer loss. These measures are actionable and can directly lower the dominant emission source.
- ****Optimize production process efficiency****: The production stage accounts for 37.7% of emissions; therefore, invest in energy-efficient technologies such as heat recovery steam generators during ammonia synthesis or switch to lower-carbon energy sources (e.g., natural gas with carbon capture, or renewable electricity). Feasibly, targeting a 10–20% reduction in production energy intensity could significantly lower the overall carbon footprint.
- ****Enhance data quality through primary data collection****: Replace unknown process contributions with actual supplier data for raw materials (e.g., phosphate rock mining, sulfuric acid production) and manufacturing energy inputs. Conduct facility-level audits to capture precise emission factors for key processes like granulation and drying. This will improve the accuracy and specificity of future LCA iterations.
- ****Evaluate distribution logistics improvements****: Although distribution is a smaller contributor (6.8%), optimize transportation routes, consolidate shipments, and use more efficient freight modes (e.g., rail over truck) to reduce fuel consumption. This pathway offers a low-cost, incremental improvement that aligns with broader sustainability goals.
- ****Implement a continuous uncertainty monitoring framework****: Given the low CV, the current model is robust, but regularly update the Monte Carlo simulations with field-measured data from use-phase trials (e.g., actual N₂O measurements). This will refine the confidence interval and support better decision-making for emission reduction investments.